Omin Kwon

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Research Interest

Computer Architectures & Machine Learning Systems, with an emphasis on co-designing ML algorithms and computer systems for efficient ML.

Education

Seoul National University Sept 2025 – present
M.S. in Computer Science and Engineering

- Advisor: Prof. Jae W. Lee, ARC Lab
- GPA: 4.3/4.3

Seoul National University Mar 2019 – Aug 2025
B.S. in Electrical and Computer Engineering (summa cum laude)

- GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)
- Leave of absence for mandatory military service (May 2021-Feb 2023)

Hansung Science High School Mar 2017 – Mar 2019
Early Graduation for Academic Excellence

Publications

SketchSSM: Write to the Full State, Read from a Compact Sketch
Omin Kwon, JoongWon Shin, Minseo Kim, Kurt Keutzer, Sehoon Kim†, Jae W. Lee†
NeurIPS 2026 LCFM Workshop; extended version under review
Project Page · † Co-corresponding authors.

HERALD: High Throughput Block Diffusion LLM Serving via CPU-GPU Cooperative KV Cache Retrieval
Omin Kwon, Doyeon Kim, Jongseok Park, Seung Yul Lee, Ion Stoica, Jae W. Lee
Under review
Paper

MAGE: All-[MASK] Block Already Knows Where to Look in Block Diffusion LLM
Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee
NeurIPS 2026; ICML 2026 AdaptFM Workshop (Best Paper Award)
Paper

STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer
Omin Kwon*, Kyungjun Oh*, Jaeyong Lee, Myungsuk Kim, Jihong Kim
ICCAD 2025
Paper · * Equal contribution. U.S. patent application with SK hynix: No. 19/569899, filed March 17, 2026.

NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs
Haeun Lee, **Omin Kwon**, Yeonhong Park, Jae W. Lee
NeurIPS 2025
Paper

AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM
KyungSoo Kim, **Omin Kwon**, Yeonhong Park, Jae W. Lee
IEEE CAL
Paper

Experiences

Teaching Assistant, Advanced Computer Architecture (4190.571) — Sept 2026 – present
Seoul National University — Seoul, South Korea

Visiting Researcher — Mar 2026 – May 2026
University of Illinois Urbana-Champaign (UIUC) — Urbana, IL, USA
- FAST Lab, hosted by Prof. Nam Sung Kim

Microelectronics (430.207B) Tutor — Mar 2025 – June 2025
Seoul National University — Seoul, South Korea

Digital System Design (430.315) Tutor — Sept 2024 – Dec 2024
Seoul National University — Seoul, South Korea

Computer Organization and Design (430.322) Tutor — Mar 2024 – June 2024
Seoul National University — Seoul, South Korea

Military Service — May 2021 – Feb 2023
Republic of Korea Air Force — South Korea

Presentations

MAGE: All-[MASK] Block Already Knows Where to Look in Block Diffusion LLM — 2026
Oral presentation at the ICML 2026 AdaptFM Workshop. arXiv.

STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer — 2025
Oral presentation at ICCAD 2025, Munich. Paper.

Honors & Awards

Best Paper Award @ ICML26 AdaptFM Workshop — July 2026
ICML 2026 AdaptFM Workshop
1 Best Paper among 123 accepted papers from 218 submissions.

Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest — Jan 2025
Ministry of Science and ICT
Designed a resolution-aware diffusion accelerator leveraging computational staleness. News: Edaily.

Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest — Dec 2024
Ministry of Education
Designed a state-aware data randomizer for high-performance SSDs, improving performance and device lifetime. News: Dong-A Ilbo.

Grand Prize, 2024 FriendliAI LLM Hackathon — July 2024
FriendliAI
Developed a knowledge-graph RAG application using the FriendliAI API.

SNU Specialized Semiconductor College Scholarship — Aug 2024 – Aug 2025
Seoul National University

Merit-Based Scholarship — 2023, 2024
Seoul National University